



## RT-ESSA Series

# Outdoor Cabinet Energy Storage System

## KEY STRENGTHS

- Support flexible expansion of PV capacity.
- Integrated EMS function, safe and stable.
- Support simultaneous access to load, battery, grid, DG, and PV.
- Support battery capacity and discharge time prediction.
- Built-in isolation transformer has strong load adaptability.

## APPLICATIONS



Off-grid area



Areas without electricity(power shortage)



Remote rural area

| Model                            | RT-ESSA0030B-0055                | RT-ESSA0050B-0055 | RT-ESSA0050B-0100 | RT-ESSA0100B-0215 |
|----------------------------------|----------------------------------|-------------------|-------------------|-------------------|
| AC data                          |                                  |                   |                   |                   |
| Rated power (kW)                 | 30                               | 50                | 50                | 100               |
| Rated voltage (V)                | 400                              |                   |                   |                   |
| Rated current (A)                | 43                               | 72                | 72                | 144               |
| Voltage range (V)                | 320V-460V                        |                   |                   |                   |
| Rated frequency (Hz)             | 50/60                            |                   |                   |                   |
| Frequency Range (Hz)             | 45-55 / 55-65                    |                   |                   |                   |
| THDi (on-grid)                   | <3%                              |                   |                   |                   |
| THDu (off-grid)                  | ≤ 1% linear; ≤ 5% non-linear     |                   |                   |                   |
| Power factor                     | 1leading~1lagging(settable)      |                   |                   |                   |
| Overload capacity                | 110% long-term                   |                   |                   |                   |
| Grid type                        | 3W+N+PE                          |                   |                   |                   |
| Isolation transformer            | 100/400                          | 200/400           | 200/400           | 270/400           |
| On / off-grid switching          | Support                          |                   |                   |                   |
| PV data                          |                                  |                   |                   |                   |
| Max. PV input voltage (V)        | 1,000                            |                   |                   |                   |
| Max. PV power (kW)               | 36/72                            | 60/120            | 60/120            | 120/180/240       |
| MPPT operating voltage range (V) | 250~850                          |                   |                   |                   |
| Buck-boost mode                  | Support                          |                   |                   |                   |
| Battery data                     |                                  |                   |                   |                   |
| Cell type                        | 3.2V/120Ah/LFP                   | 3.2V/120Ah/LFP    | 3.2V/280Ah/LFP    | 3.2V/280Ah/LFP    |
| Rated voltage (V)                | 460.8V, 1P144S                   | 460.8V, 1P144S    | 358.4V, 1P112S    | 768V, 1P240S      |
| Rated energy (kWh)               | 55.296                           | 55.296            | 100.352           | 215.04            |
| Operating voltage range (V)      | 403.2~511.2                      | 403.2~511.2       | 319.2~397.6       | 672~850           |
| Max. charge and discharge rate   | 1C@25°C                          | 1C@25°C           | 0.5C@25°C         | 0.5C@25°C         |
| Number of battery cycles         | ≥3,800                           | ≥3,800            | ≥6,000            | ≥6,000            |
| General data                     |                                  |                   |                   |                   |
| Dimensions W *D *H (mm)          | 1,780×1,520×2,290                | 1,780×1,520×2,290 | 1,780×1,585×2,290 | 2,460×1,560×2,425 |
| Weight (kg)                      | 2,000                            | 2,200             | 2,400             | 3,900             |
| Operating temperature (°C)       | 0 ~ +45                          |                   |                   |                   |
| Relative humidity                | 0 ~95% (non-condensing)          |                   |                   |                   |
| Ingress protection               | IP54                             |                   |                   |                   |
| Noise emission (dB)              | <75                              |                   |                   |                   |
| Max. operating altitude (m)      | 5,000(>3,000 derating)           |                   |                   |                   |
| Cooling                          | Intelligent air cooling          |                   |                   |                   |
| Fire extinguishing system        | Aerosols/NOVEC1230(optional)     |                   |                   |                   |
| Display                          | Touch LCD display+cloud platform |                   |                   |                   |
| EMS communication                | RS485, TCP/IP                    |                   |                   |                   |